INDIAN INSTITUTE OF TECHNOLOGY PALAKKAD
Department of Mechanical Engineering

ME5610 SOFT COMPUTING (TEST 2)

Duration: 50 Minutes **Max Marks: 20**

Note:

- **Answer all the questions in the answer sheet only. .**
- **Draw neat schematic diagrams wherever required.**
- **Any doubts related to the question paper will not be addressed by the invigilators/ course instructor(s) during the examination.**
- **Assume suitable missing data (if any) and mention it in the answer sheet.**
- **Do not write anything on the question paper.**

1. To demonstrate a fuzzy logic controller the following example is considered. Tip at a restaurant is decided (from 0 to 25 %) based on the service (rated in points from 0 to 10) and food quality (rated in the points from 0 to 10) by a person. The membership functions of the input parameters service, and food quality are shown in Figure 1 and Figure 2 respectively. The membership functions for the output parameter (Tip) are shown in Figure 3.

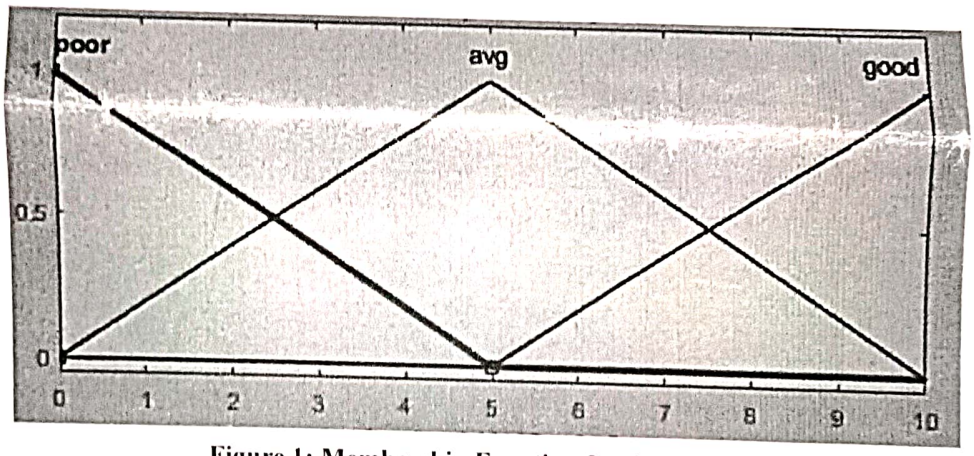

Figure 1: Membership Function for food quality

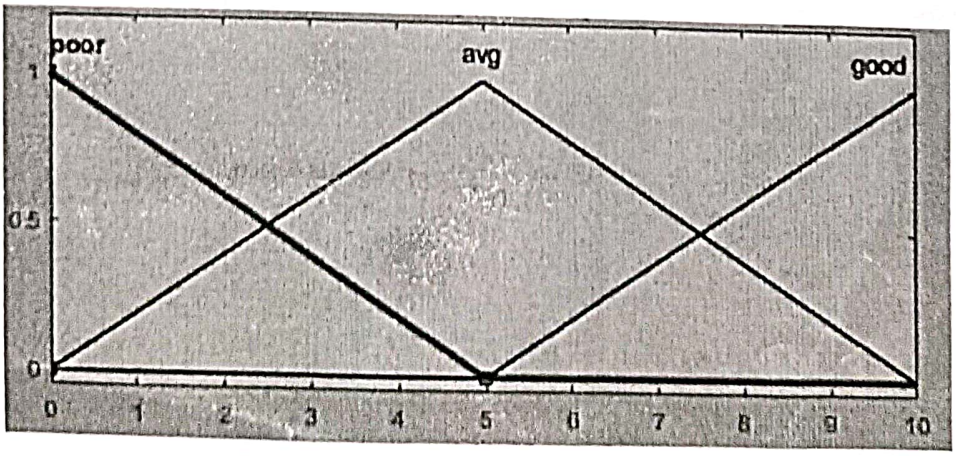

Figure 2: Membership function for service

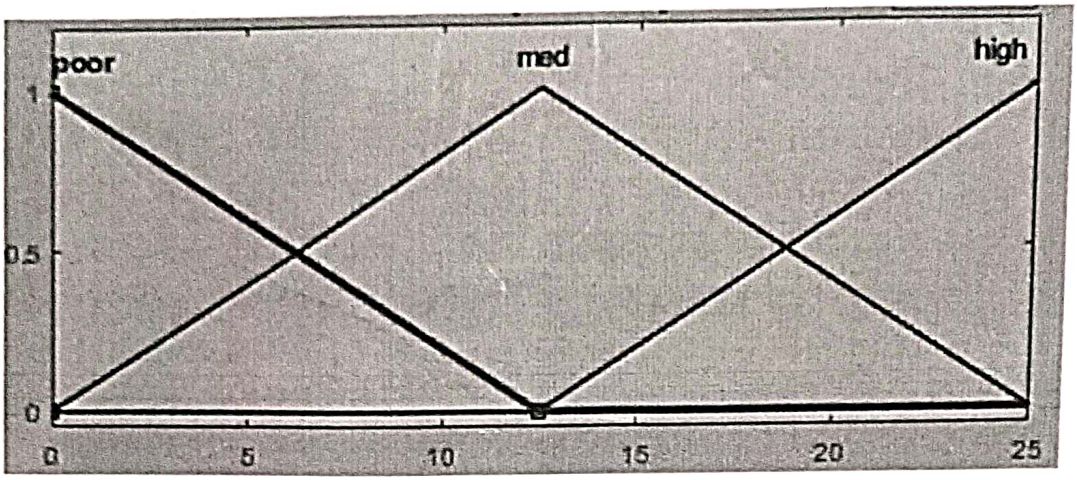

Figure 3: Membership functions for Tip percentage

(i) Find the firing strength (α value) for the rule mentioned below as per the given inputs:

Rule 1: If *food quality* is Poor **AND** *service* is Poor then *Tip* is Poor

(Input rated points given by the person are food quality=1; service= 2. For this Output Tip = 2%)

$\mu = ?$ $\mu = ?$

(ii) Find the firing strength (α values) for the rule mentioned below as per the given inputs:

Rule 2: If *food quality* is Good **OR** *service* is Good then *Tip* is High

(Input rated points given by the person are food quality=9; service= 8. For this Output Tip= 22%)

[4 Marks]

(Question 2 and 3: 3 Marks each)

2. Give the mathematical and graphical representations of the log-sigmoid and tan-sigmoid functions. (No need to explain the activation functions.. Just mention the mathematical and graphical representations).

3. If A(x) and B(x) are two fuzzy sets. Determine inaccuracy of the fuzzy set B (x) with respect to the fuzzy set A (x).

$A(x) = \{(x_1, 0.3), (x_2, 0.4), (x_3, 0.3), (x_4, 0.2)\}$

$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8), (x_4, 0.9)\}$

(Question 5 to 8: 0.5 Mark each)

4. If A and B are two crisp sets in the universal set X, according to law of associativity, $(A \cup B) \cup C =$ _____ ?
5. If A and B are two crisp sets in the universal set X, according to the law of commutativity what is the relation between the two crisp sets A and B?
6. If A is a crisp set in the universal set X, according to the law of identity $A \cap X =$ _____ ?
7. If A and B are two crisp sets, according to De Morgan's Law $\overline{(A \cup B)} =$ _____ .

(Question 9 to 12: 2 Mark each)

8. If A(x) and B(x) are two fuzzy sets, then find the bounded difference of the two fuzzy sets.

   A(x) = {(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.2)} B(x) = {(x_1, 0.5), (x_2, 0.7), (x_3, 0.8), (x_4, 0.3)}

9. If A(x) and B(y) are two fuzzy sets, then find the cartesian product of the two sets A(x) and B(x):

   A(x) = {(x_1, 0.4), (x_2, 0.3), (x_3, 0.5), (x_4, 0.8)} B(y) = {(x_1, 0.8), (x_2, 0.4), (x_3, 0.3)}

10. If A and B are two fuzzy sets. Find the composition of two fuzzy sets A and B.

$$A = \begin{bmatrix} 0.1 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \qquad B = \begin{bmatrix} 0.2 & 0.5 & 0.8 \\ 0.1 & 0.7 & 0.5 \end{bmatrix}$$

11. Develop a single node artificial neural network (ANN) architecture that functions as an OR gate, specifying its weightages, bias values. Demonstrate it for all truth table of the OR gate.